

Knee Osteoarthritis Severity Grading

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Contents

Abstract	2
1. Introduction	2
1.1 Background and Related Work	3
1.2 Hypothesis and Assumptions	3
2. Methods	3
2.1 Dataset and Preprocessing	4
2.2 Model Architectures	5
2.3 Ordinal Classification Approach	8
3. System Interface	8
4. Results	10
4.1 Performance metrics	10
4.2 Discussion	12
5. Limitations	13
6. Future Improvements	14
7. Conclusion	14
References	15

Abstract

Knee Osteoarthritis (OA) severity is commonly assessed through manual analysis of X-ray images, a process that can be time-intensive and subjective, especially when dealing with large datasets. In this project, automated models were developed to address these limitations, including both image-only approaches and multimodal models that integrate radiographic images with clinical metadata. Architectures such as ResNet and Swin Transformer are used for image feature extraction, while clinical features including age, BMI, and radiographic indicators enhance prediction performance. The proposed approach also accounts for the ordinal nature of OA severity grades. Experimental results show that multimodal models outperform traditional image-only models, achieving higher macro F1-score and balanced accuracy. The system further groups patients into risk levels, making it suitable for practical clinical use.

1. Introduction

Knee Osteoarthritis (OA) is a degenerative condition and one of the most common causes of disability worldwide, especially in older populations. The disease involves progressive cartilage breakdown, narrowing of the joint space, and structural changes in bone, which together result in pain and limited movement. Detecting the condition early and accurately determining its severity are essential for guiding treatment and slowing disease progression.

Traditionally, OA severity is assessed using radiographic images and graded according to the Kellgren–Lawrence (KL) scale. However, this process is manual, time-intensive, and often inconsistent due to subjective interpretation by clinicians. These challenges highlight the need for more consistent and scalable automated methods to support diagnosis.

Recent advancements in deep learning, particularly convolutional neural network (CNN) and transformer-based architectures, have shown strong performance in medical image analysis. However, many existing approaches rely solely on image data and overlook valuable clinical metadata that can provide additional context for more accurate predictions. Incorporating both image and clinical data is expected to improve performance by capturing supplementary information. Additionally, OA severity follows a progressive structure rather than independent classes; modeling it as an ordinal task helps preserve relationships between adjacent grades and reduces large misclassification errors.

To address this gap, this project develops a multimodal deep learning framework that combines knee X-ray images with clinical features to improve OA severity classification. Different architectures, including ResNet and Swin Transformer, are explored to extract meaningful representations from the data. The models are evaluated using performance measures such as macro F1-score, balanced accuracy, and confusion matrix.

1.1 Background and Related Work

Several studies have explored automated approaches for classifying osteoarthritis using deep learning methods. CNN-based architectures, particularly architectures such as ResNet, have been commonly applied due to their ability to learn detailed spatial features from medical images and their effectiveness in large-scale tasks. More recently, transformer-based models like Swin Transformer have demonstrated improved capability in capturing broader image context.

While image-based models have achieved significant success, they often ignore clinical data such as patient demographics and radiographic measurements. Some recent research has started to explore multimodal approaches by combining image and structured clinical data, showing improved performance over unimodal approaches.

Despite these advancements, several challenges persist. Class imbalance is a significant issue, as certain severity levels are underrepresented in the dataset and can bias model training. In addition, as OA progresses gradually, adjacent classes tend to overlap, making them difficult to distinguish, particularly for intermediate grades where subtle differences increase misclassification. This points to the need for models that can handle both class imbalance and the ordered structure of severity while making use of multiple data sources.

1.2 Hypothesis and Assumptions

This project is based on main hypotheses related to knee osteoarthritis (OA) classification. The first hypothesis is that combining radiographic images with clinical metadata leads to improved predictive performance compared to image-only models, as clinical variables provide additional context to structural features captured in X-rays. The second is that OA severity is better represented as an ordinal problem rather than a standard multi-class classification since the grades represent a natural progression of disease severity.

In addition to these hypotheses, several assumptions are considered. The class distribution is imbalanced, which can bias model learning and therefore requires mitigation through techniques such as weighted loss functions. Furthermore, overlap between adjacent classes is anticipated due to the gradual nature of OA progression, making certain classes more difficult to distinguish.

2. Methods

This project develops a multimodal deep learning framework for the classification of knee osteoarthritis (OA) severity by combining X-ray image data with clinical metadata. The goal is to design and evaluate how distinctive features contribute to model performance and to determine whether combining them leads to more reliable predictions. The approach

includes three main stages: data preparation, model development, and an ordinal classification strategy that reflects the progressive nature of OA.

Three model setups are evaluated: image-only, metadata-only, and multimodal. This allows for a structured comparison and provides a clear understanding of how each type of input contributes to the final performance.

2.1 Dataset and Preprocessing

The dataset consists of knee X-ray images categorized into Kellgren–Lawrence (KL) grades, along with associated clinical metadata. Each sample is identified using a patient ID and knee side (left or right), enabling accurate alignment between images and corresponding clinical features.

The dataset is organized into training, validation, and test sets based on directory structure, where each folder represents a KL grade class. For each image, a patient ID and knee side (left or right) are extracted from the filename. A structured DataFrame is created containing image paths, labels, and identifiers.

To prevent data leakage, patient IDs are explicitly checked across training and test sets. Any overlap is removed to ensure that the model is evaluated only on unseen patients, preserving the validity of performance metrics.

Clinical metadata is obtained from multiple structured sources and merged using patient ID and knee side as keys. The metadata includes features such as age, BMI, gender, race, knee pain scores, and radiographic indicators (e.g., joint space narrowing and osteophyte formation). Numerical features are normalized prior to model input; RobustScaler is applied in the metadata-only and ResNet-based multimodal models, which is less sensitive to outliers, while StandardScaler is used in the Swin-based multimodal models. Missing values are handled using median imputation. Categorical variables such as side, gender, race, and ethnicity are encoded numerically and used as direct inputs to the model. An additional derived feature representing overall knee pain is computed by combining left and right knee pain scores.

For image preprocessing, all X-ray images are resized to 224×224 pixels, to ensure compatibility with models, and normalized using ImageNet statistics. To improve generalization and reduce overfitting, data augmentation techniques such as horizontal flipping and slight rotations are applied during training.

2.2 Models Architectures

Three categories of models are implemented to evaluate the contribution of different data types: image-only models, metadata-only models, and multimodal models. Within the image-only and multimodal settings, both architecture-level comparisons (ResNet vs Swin) and learning formulation comparisons (multi-class vs ordinal) are explored.

i. Image-Only Models

Image-based models take X-ray images as input and learn feature representations directly from visual data. Two primary architectures Resnet34 and Swin Transformer (Swin-Tiny) are explored, both initialized with ImageNet pretrained weights.

Resnet: The ResNet34 architecture is a convolutional neural network that utilizes residual connections to enable deep feature extraction without degradation in training performance. The convolutional backbone captures spatial patterns such as edges, textures, and structural abnormalities in the knee joint. In this implementation, the final classification layer is replaced with a task-specific head consisting of a dropout layer followed by a fully connected layer that outputs predictions for all KL grades.

Swin Transformer: The Swin Transformer is a hierarchical vision transformer that processes images using shifted window attention. This design enables efficient modeling of both local features and long-range dependencies, which is particularly useful for capturing subtle structural variations in OA progression. The classification head is modified by replacing the original head with a LayerNorm layer, followed by dropout (0.3) and a linear layer.

For multi class classification the output layer produces K logits corresponding to KL grades, followed by a Softmax activation to obtain class probabilities. The models are trained using cross-entropy loss, which treats each class independently. To mitigate class imbalance, class weights and weighted random sampling are applied during training, ensuring that minority classes are sufficiently represented. Additional manual weight multipliers are applied on top of the balanced class weights (e.g., 1.20× for KL0 and KL1, 1.25× for KL2) to further emphasize underrepresented grades. Training follows a two-phase strategy: the backbone is frozen for the first five epochs while only the classification head is trained, after which the full network is unfrozen and trained end-to-end. Separate learning rates are used, with 3e-5 for the backbone and 3e-4 for the head. Optimization is performed using the AdamW optimizer and a CosineAnnealingLR scheduler is used to improve convergence.

To account for the progressive nature of OA severity, both architectures are extended to an ordinal classification framework. Instead of predicting K independent classes, the models output K-1 logits, where each output represents whether the severity exceeds a particular

threshold. Labels are transformed into cumulative targets, and predictions are obtained by applying sigmoid activation followed by threshold-based decoding. The loss function combines binary cross-entropy with a monotonicity constraint, which penalizes violations of the ordinal structure and enforces consistent ordering of outputs.

This setup enables direct comparison between ResNet vs Swin (architecture) and multi-class vs ordinal (learning formulation), while keeping training conditions consistent.

ii. Metadata-Only Models

The metadata-only model evaluates the predictive capability of structured clinical features independently of image data. The input includes demographic variables (age, BMI, gender, race), pain scores, and radiographic indicators such as joint space narrowing and osteophyte formation.

The model is implemented as a fully connected neural network with two hidden layers (256 and 128 units), each followed by ReLU activation and batch normalization. Input features are normalized using RobustScaler, and missing values are handled using median imputation.

The model follows an ordinal classification approach, where labels are encoded into cumulative form. The loss function combines binary cross-entropy with class-specific weighting and a monotonicity constraint. Predictions are generated using threshold-based decoding.

iii. Multimodal Models

Multimodal models integrate image, and metadata features to leverage complementary information and improve classification performance.

Naive Multimodal Approach

A naive multimodal baseline is implemented by converting clinical metadata into an image representation and appending it directly to the X-ray image. Specifically, numerical and categorical metadata are first scaled and concatenated into a single feature vector. This vector is then tiled into a narrow strip, normalized, resized to a fixed height, and converted into an RGB image. The metadata strip is placed below the resized X-ray image to form one combined input image, which is then used for training. This design allows the model to learn jointly from radiographic and clinical information without requiring a separate metadata branch. The combined image is processed using a pretrained ResNet18 backbone. The original classification head is replaced with a linear layer matching the number of KL classes, and the model is trained as a standard multi-class classifier using cross-entropy loss.

Resnet + Metadata Approach

In the multi-class multimodal configuration, feature representations are extracted separately from image and metadata inputs. The image branch uses a ResNet34 backbone, where the final classification layer is removed and replaced with a projection layer to obtain a compact feature representation. The metadata branch processes structured inputs using embedding layers for categorical variables and dense layers for numerical features, producing a metadata feature vector. The two representations are concatenated to form a unified feature vector, which is passed through fully connected layers for classification. The model is trained using Softmax activation and cross-entropy loss. This model allows evaluation of multimodal learning without enforcing ordering constraints.

To incorporate the ordered nature of OA severity, the ResNet-based multimodal model is extended to an ordinal formulation. The output layer is modified to produce $K-1$ logits, and labels are encoded into cumulative targets. The model is trained using binary cross-entropy with class-specific weights and a monotonicity constraint to ensure consistent ordering of predictions. Threshold-based decoding is used to obtain the final KL grade.

Swin + Metadata approach

The Swin-based multimodal model combines image and clinical features through feature-level fusion. The image branch uses a pretrained Swin-Tiny backbone with the classification head removed to extract image representations.

The metadata branch processes structured clinical inputs using a small neural network with two fully connected layers (128 units each), each with ReLU activation, producing a compact feature embedding. Metadata is scaled using StandardScaler before input. The concatenated image and metadata features are then passed through a 256-unit fully connected layer with ReLU activation and 0.4 dropout before the final output layer. The image and metadata embeddings are concatenated and passed through fully connected layers for classification.

In the multi-class setting, the model uses Softmax activation and cross-entropy loss, with class imbalance handled through class weights and a weighted sampler. In the ordinal variant, the final layer is modified to output $K-1$ logits, and labels are encoded as cumulative targets. The model is trained using binary cross-entropy with a monotonicity constraint, and predictions are obtained using threshold-based decoding.

This setup enables direct comparison between multi-class and ordinal formulations while keeping the architecture unchanged.

2.3 Ordinal Classification Approach

Knee osteoarthritis (OA) severity is modeled as an ordinal problem, as the Kellgren–Lawrence (KL) grades represent a progressive scale rather than independent classes. To capture this structure, selected models are formulated using an ordinal classification framework instead of standard multi-class classification.

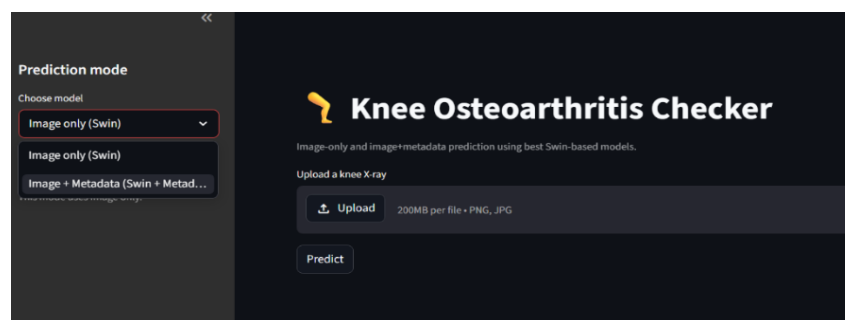
In this approach, a problem with K classes is transformed into $K-1$ binary classification tasks. Each output predicts whether the severity exceeds a given threshold (e.g., $KL > 0, KL > 1, \dots$). Ground truth labels are converted into cumulative binary targets accordingly. The models output $K-1$ logits, which are trained using binary cross-entropy with logits. To enforce the ordinal structure, a monotonicity constraint is added to the loss function, penalizing cases where higher-order logits exceed lower-order ones. This ensures consistency in predictions across ordered thresholds.

During inference, sigmoid activation is applied to the logits, and fixed thresholds are used to determine whether each condition is satisfied. In image-only ordinal models, thresholds are evenly spaced in the range 0.45 to 0.60 using `torch.linspace`, while in multimodal ordinal models fixed values of [0.45, 0.50, 0.55, 0.60] are applied. The final KL grade is obtained by counting the number of thresholds exceeded.

For comparison, models are also trained using standard multi-class classification with Softmax and cross-entropy loss. This allows evaluation of whether explicitly modeling the ordinal structure leads to improved performance.

3. System Interface

A user interface is developed using Streamlit to provide an accessible front end for the proposed OA grading system. The interface supports two prediction modes: an image-only mode using the best-performing Swin Transformer model, and a multimodal mode using the Swin + Metadata (Ordinal) model. Users can select the desired mode from a dropdown menu. In the image-only mode, only an X-ray image upload is required. In the multimodal mode, the system automatically reads patient metadata from the uploaded image filename and populates clinical fields such as age, BMI, gender, and radiographic features.



Knee Osteoarthritis Checker – Dropdown selection for Image only prediction and Image + Metadata multimodal mode

Once the inputs are provided, the system processes the data through the trained model and generates a predicted KL grade. The output is displayed along with confidence scores, providing a clear interpretation of the model's prediction.

The interface also categorizes predictions into risk levels (Low, Moderate, High, and Severe) and generates a personalized health education summary for the predicted KL grade. This includes grade-specific guidance on recommended exercises, dietary advice, activities to avoid, and when to seek medical help. The system displays a clear disclaimer that this output is for general education only and does not constitute a medical diagnosis. This design demonstrates how the proposed model can be applied in a practical setting for automated and efficient OA assessment.

Prediction mode

Choose model
Image + Metadata (Swini + ...)

Metadata
Metadata will be auto-filled from patient metadata using the uploaded image.

Metadata
Values loaded from patient metadata.

Age
70.0

Gender
Female

BMI
24.2

Right knee pain
2.0

Left knee pain
0.0

Weight (kg)
73.3

Height (cm)
1739.0

Overall knee pain
1.0

JSN medial
0.0

JSN lateral
0.0

Osteo femur medial
0.0

Osteo tibia lateral
0.0

Sclerosis femur medial
0.0

Knee Osteoarthritis Checker
Image-only and image+metadata prediction using best Swini-based models.

Upload a knee X-ray
9821149_2.png (22.44B)

Predict

Prediction

KL Grade
0

Confidence
96.9%

Risk
Low Risk

Stage: **No OA**
Base risk: **Low Risk**
Final risk: **Low Risk**

This is a general education summary, not a medical diagnosis.

Summary

- Your model output suggests KL Grade 0 (early stage).
- This is a general education summary, not a medical diagnosis.
- The focus is usually on prevention, healthy movement, and protecting the joint.

Exercise

- Low-impact walking in short comfortable sessions
- Gentle range-of-motion exercises
- Light strengthening for the thigh and hip muscles
- Regular movement to keep the joint active

Diet

- Aim for balanced meals with vegetables, fruits, and enough protein
- If BMI is elevated, gradual weight loss may reduce knee stress
- Limit highly processed foods and sugary drinks
- Stay hydrated

Avoid

- High-impact running or jumping
- Long periods of sitting without movement
- Deep squats if they cause discomfort
- Heavy lifting without guidance

When to seek medical help

- See a clinician if pain is worsening, swelling is increasing, or walking becomes difficult
- Seek medical advice if you develop sudden severe pain or new instability

> Model output details

> Risk inputs

This information is for general education only and is not a medical diagnosis. Please consult a qualified clinician for personal medical advice.

Figure shows Interface prediction output showing KL Grade, confidence score, risk level, and health education summary panel with exercise, diet, and avoidance guidance

4. Results

The proposed framework was evaluated using multiple model configurations, including image-only, metadata-only, and multimodal approaches.

4.1 Performance metrics

Performance was assessed using macro F1-score, balanced accuracy, area under the ROC curve (AUC), and confusion matrix analysis. These metrics provide a comprehensive assessment, particularly under class imbalance conditions.

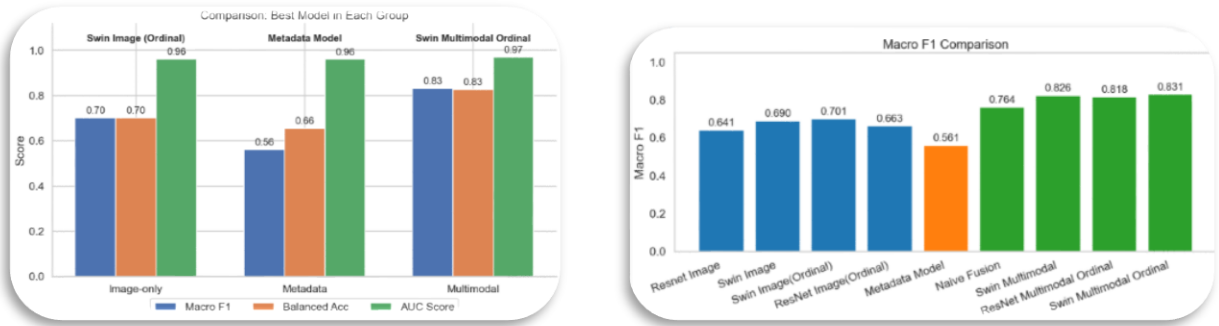


Table 1 summarizes the performance of all nine model configurations across three metrics. All models were trained with a batch size of 32, up to 40 epochs with early stopping (patience = 12), and optimized using AdamW.

Model	Pipeline	Macro F1	Bal. Acc	AUC
ResNet	Image-only	0.6408	0.6355	0.8631
Swin	Image-only	0.6901	0.7020	0.9059
Swin Ordinal	Image-only	0.7015	0.7011	0.9615
ResNet Ordinal	Image-only	0.6633	0.6500	0.9461
Meta	Metadata	0.5610	0.6557	0.9606
Naive	Multimodal	0.7637	0.7644	0.9278
ResNet + Meta Ordinal	Multimodal	0.8175	0.8115	0.9724
Swin + Meta	Multimodal	0.8258	0.8237	0.9354
Swin + Meta Ordinal	Multimodal	0.8312 ★	0.8269	0.9702

Table 1: Performance of all model configurations. ★ = best overall model.

Among image-only models, the Swin Transformer outperforms ResNet, achieving a macro F1-score of 0.6901 compared to 0.6408, along with higher balanced accuracy (0.7020 vs 0.6355) and AUC (0.9059 vs 0.8631). This indicates that transformer-based architectures are more effective at capturing contextual patterns in knee X-ray images.

The metadata-only model achieves a macro F1-score of 0.5610, with relatively high AUC (0.9606) but lower classification performance. This suggests that while metadata contains useful ranking information, it is insufficient for accurate class separation when used independently.

The naive multimodal approach improves performance to 0.7637 F1-score, demonstrating that incorporating clinical information is beneficial even with simple fusion. Structured multimodal models achieve the highest performance. The ResNet + Metadata Ordinal model achieves 0.8175 F1-score, while the Swin + Metadata model improves further to 0.8258, confirming the advantage of combining strong feature extraction with multimodal integration.

Ordinal Classification

The impact of ordinal classification is evident across both architectures. The Swin-based ordinal model achieves 0.7015 F1-score, outperforming its multi-class counterpart (0.6901). Similarly, the ResNet ordinal model improves over the standard version (0.6633 vs 0.6408).

The most significant improvement is observed in the multimodal setting. The Swin + Metadata (Ordinal) model achieves the best overall performance with a macro F1-score of 0.8312, balanced accuracy of 0.8269, and AUC of 0.9702. This demonstrates that incorporating ordinal constraints leads to more consistent and accurate predictions.

Pipeline-Level Comparison

A comparison across pipelines highlights the contribution of each data type:

- Image-only pipeline: Best model is Swin Ordinal with 0.7015 F1-score
- Metadata pipeline: Best model achieves 0.5610 F1-score
- Multimodal pipeline: Best model (Swin+Meta Ordinal) achieves 0.8312 F1-score

These results confirm that multimodal learning provides the largest performance gain, significantly outperforming both image-only and metadata-only approaches.

Normalized Confusion Matrices																	
Image-only Best: Swin Image (Ordinal)						Metadata Best: Metadata Model						Multimodal Best: Swin Multimodal Ordinal					
Actual	KL0	KL1	KL2	KL3	KL4	Actual	KL0	KL1	KL2	KL3	KL4	Actual	KL0	KL1	KL2	KL3	KL4
	0.71	0.27	0.02	0.00	0.00		1.00	0.00	0.00	0.00	0.00		0.87	0.13	0.00	0.00	0.00
	0.29	0.52	0.18	0.01	0.00		0.53	0.29	0.18	0.00	0.00		0.32	0.57	0.11	0.00	0.00
	0.05	0.25	0.65	0.05	0.00		0.12	0.13	0.65	0.09	0.01		0.05	0.24	0.70	0.00	0.00
	0.00	0.02	0.16	0.78	0.04		0.00	0.00	0.00	0.34	0.66		0.00	0.00	0.00	0.99	0.00
	0.00	0.00	0.00	0.16	0.84		0.00	0.00	0.00	0.00	1.00		0.00	0.00	0.00	0.00	1.00
Predicted					Predicted					Predicted							

Confusion matrices provide detailed insight into class-wise performance. A consistent observation is that KL1 is the most difficult class to predict, with lower true positive rates compared to other classes. In the best multimodal model (Swin+Meta Ordinal), KL1 achieves a true positive rate of only 0.57, with 0.32 of KL1 samples misclassified as KL0. Most misclassifications occur between KL0, KL1, and KL2, indicating strong overlap in early-stage OA. These classes are closely related, and errors in one class often affect predictions of neighboring classes. In contrast, higher severity classes such as KL3 and KL4 show strong diagonal dominance, with KL3 achieving 0.99 and KL4 achieving 1.00 true positive rates in the best multimodal model, indicating high classification accuracy due to more distinct structural changes.

4.2 Discussion

The results demonstrate the impact of model architecture, data integration, and learning formulation on OA severity classification.

First, the comparison between ResNet and Swin highlights the importance of architecture. The Swin Transformer consistently outperforms ResNet across both multi-class and ordinal settings, suggesting that its ability to capture global context is beneficial for medical image analysis.

Second, multimodal learning provides the most significant improvement. While image-only models achieve moderate performance, and metadata-only models show limited classification ability, combining both leads to substantial gains. The improvement from 0.7015 (image-only) to 0.8312 (multimodal) highlights the importance of integrating complementary information. The best multimodal multi-class model (Swin+Meta, F1=0.8258) improves over the best image-only multi-class model (Swin, F1=0.6901) by +0.1357 in macro F1 and +0.1217 in balanced accuracy (0.8237 vs 0.7020), representing a 19.7% relative improvement in F1-score. While the best multimodal ordinal model (Swin+Meta Ordinal, F1=0.8312) improves over the best image-only ordinal model (Swin

Ordinal, $F1=0.7015$) by $+0.1297$ in macro F1 and $+0.1258$ in balanced accuracy (0.8269 vs 0.7011), representing an 18.5% relative improvement in F1-score.

Third, ordinal classification improves performance across all model types. The consistent improvement in F1-score for both ResNet and Swin indicates that explicitly modeling the ordered structure of OA severity reduces large classification errors and improves prediction consistency.

Despite these improvements, confusion matrix analysis reveals persistent challenges. The model consistently struggles with KL1, which shows lower accuracy due to subtle and overlapping features. Additionally, KL0, KL1, and KL2 are strongly interdependent, leading to frequent misclassification among these classes. This reflects the gradual progression of OA rather than clearly separable categories.

In contrast, higher severity grades (KL3 and KL4) are predicted with significantly higher accuracy due to more pronounced structural changes. The results validate the proposed framework, demonstrating that combining multimodal learning with ordinal classification and transformer-based architectures yields the best performance.

5. Limitations

Despite the strong performance of the proposed framework, several limitations remain. First, the dataset exhibits class imbalance, with a higher number of samples in lower severity grades compared to higher grades. Although techniques such as weighted sampling and loss weighting are applied, imbalance still affects model performance, particularly for minority classes.

Second, there is significant overlap between early-stage classes, especially KL0, KL1, and KL2. These classes share similar structural characteristics in X-ray images, making them difficult to distinguish even for advanced models. This limitation is reflected in the confusion matrix, where most misclassifications occur among these classes.

Third, while ordinal classification improves performance, it does not completely eliminate errors in intermediate grades. The effectiveness of ordinal modeling may depend on the quality and separability of features within the dataset.

Additionally, the models are trained and evaluated on a specific dataset, which may limit generalizability to other populations, imaging conditions, or clinical settings. Variations in image quality and acquisition protocols could affect performance in real-world deployment.

Finally, although a user interface is developed, the system has not been validated in a clinical environment, and its practical applicability requires further evaluation.

6. Future Improvements

Several directions can be explored to further improve the proposed system. One potential improvement is extending ordinal classification to all model configurations, including earlier multi-class variants, to fully evaluate its impact across architectures. Additionally, more advanced ordinal methods or adaptive thresholding strategies could be explored to further improve prediction consistency.

Another important direction is to investigate why clinical metadata significantly impacts model performance. While the current results show that metadata improves classification, a deeper analysis of feature importance and contribution could provide insights into which variables are most influential and how they interact with image features.

Further improvements can be made by enhancing the set of clinical features used in the model. Incorporating additional relevant factors such as history of past knee injury, hereditary or genetic predisposition, and other clinical indicators may provide richer context and improve model performance, particularly for early-stage OA detection.

From a modeling perspective, more advanced multimodal fusion techniques can be explored. While feature concatenation is effective, approaches such as attention-based fusion or dynamic feature weighting could allow better interaction between image and metadata features.

Expanding the dataset is also critical. Increasing both the size and diversity of data would improve generalization and reduce bias caused by class imbalance. This is especially important for improving performance on underrepresented classes.

From a practical standpoint, the system can be extended for real-time deployment and integration into clinical workflows. Additionally, improving model interpretability, for example through visualization techniques, could increase trust and usability in real-world applications.

7. Conclusion

This project presents a multimodal deep learning framework for knee osteoarthritis severity classification by integrating radiographic images with clinical metadata. Multiple model configurations are explored, including image-only, metadata-only, and multimodal approaches, along with both multi-class and ordinal learning formulations.

The results demonstrate that multimodal models significantly outperform unimodal approaches, highlighting the importance of combining complementary data sources. Among all models, the Swin + Metadata (Ordinal) model achieves the best performance, confirming the effectiveness of both transformer-based architectures and ordinal

classification. The study also shows that ordinal modeling improves prediction consistency by incorporating the progressive nature of OA severity. These results validate the initial hypotheses, demonstrating that multimodal learning improves performance over unimodal approaches, and that ordinal classification further enhances prediction consistency by modeling the ordered nature of OA severity. However, challenges remain in distinguishing early-stage classes due to overlapping features.

Overall, the proposed framework provides an effective and scalable approach for automated OA severity assessment. With further improvements and validation, it has the potential to support clinical decision-making and reduce reliance on manual interpretation.

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